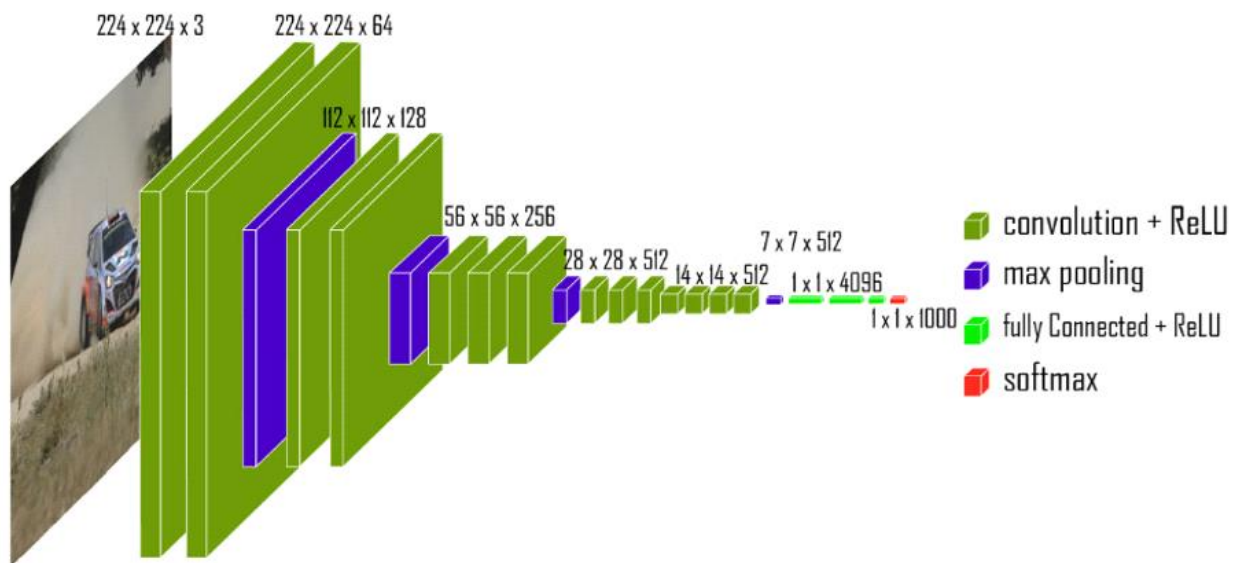


딥러닝 영상처리- Lecture Week10-1

VGG(Visual Geometry Group), 2014)



- 2014 ImageNet Challenge 2위 모델
- GoogleNet(2014 ImageNet Challenge 1위) 모델에 비해 구조가 단순하고, 성능도 큰 차이가 없어서 유명
- 모델의 깊이(레이어 수)에 따라서 VGG11, VGG13, VGG16, VGG19 로 구분
- 모델의 깊이가 어떤 영향을 주는지 연구하기 위해 설계된 CNN

Table 1: **ConvNet configurations** (shown in columns). The depth of the configurations increases from the left (A) to the right (E), as more layers are added (the added layers are shown in bold). The convolutional layer parameters are denoted as “conv(receptive field size)-(number of channels)”. The ReLU activation function is not shown for brevity.

ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224 × 224 RGB image)					
conv3-64	conv3-64 LRN	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 conv1-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256 conv3-256
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

Table 2: **Number of parameters** (in millions).

Network	A,A-LRN	B	C	D	E
Number of parameters	133	133	134	138	144

VGG 특징

<<https://arxiv.org/abs/1409.1556>>

1. 3x3 크기 필터 사용
2. ReLU 함수의 적극적 활용
3. 학습 파라미터 수 감소

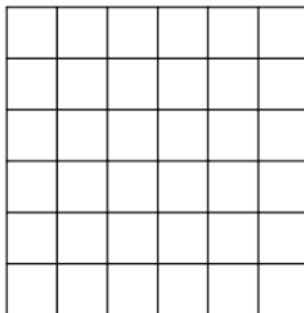
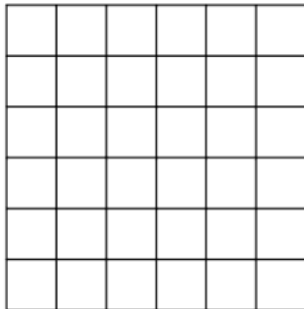
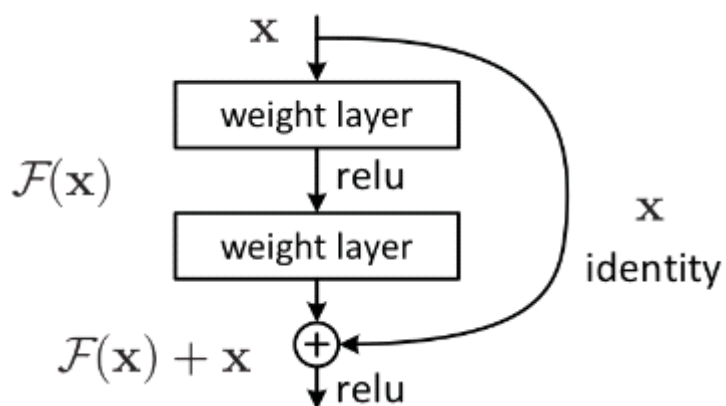
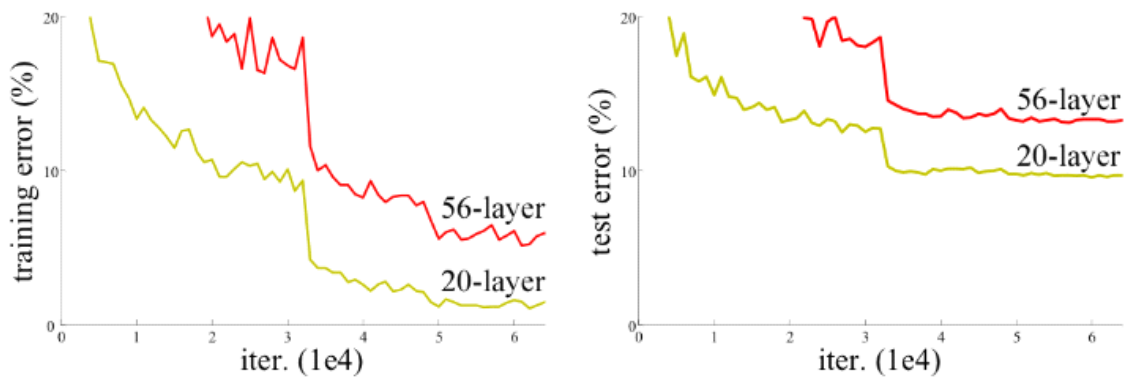


Table 3: **ConvNet performance at a single test scale.**

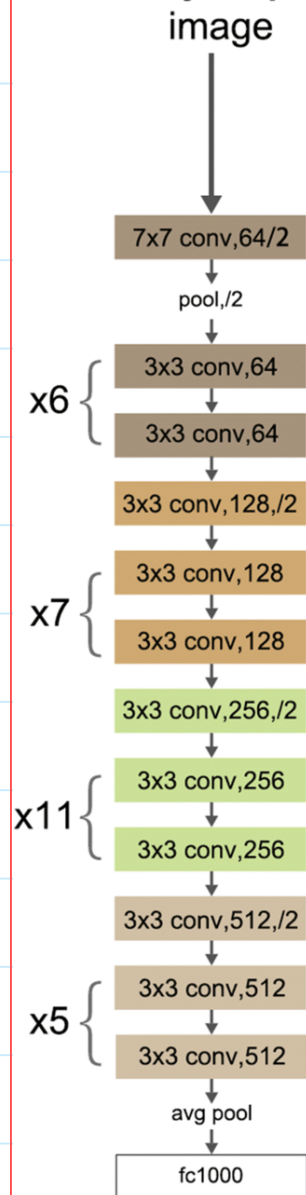
ConvNet config. (Table 1)	smallest image side		top-1 val. error (%)	top-5 val. error (%)
	train (S)	test (Q)		
A	256	256	29.6	10.4
A-LRN	256	256	29.7	10.5
B	256	256	28.7	9.9
C	256	256	28.1	9.4
	384	384	28.1	9.3
	[256;512]	384	27.3	8.8
D	256	256	27.0	8.8
	384	384	26.8	8.7
	[256;512]	384	25.6	8.1
E	256	256	27.3	9.0
	384	384	26.9	8.7
	[256;512]	384	25.5	8.0

ResNet

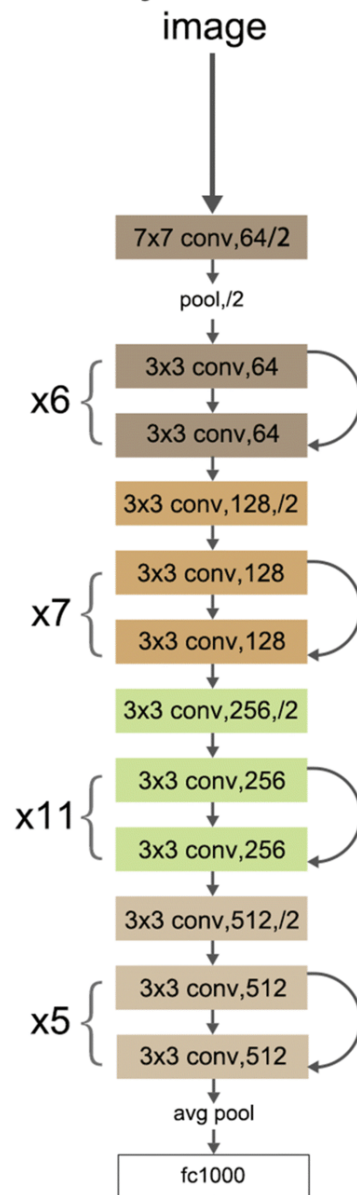
- ImageNet Challenge 2015 우승 모델
- Residual Block 이라는 이름의 모듈을 깊게 쌓음
(가장 깊게 쌓은 모델은 152개의 레이어 수 --> VGG-19보다 8배)
- 레이어를 깊게 쌓았음에도 적은 수의 파라미터를 바탕으로 설계
- 레이어를 깊게 쌓을 때 발생할 수 있는 문제(Overfitting, Degradation, 계산비용 증가, 학습 시간 증가, Gradient Vanishing) 해결



34-layer plain



34-layer residual



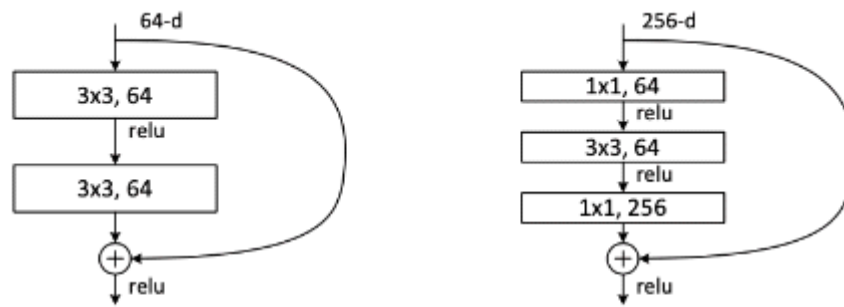


Figure 5. A deeper residual function $\mathcal{F}$ for ImageNet. Left: a building block (on 56×56 feature maps) as in Fig. 3 for ResNet-34. Right: a “bottleneck” building block for ResNet-50/101/152.

